

# The empirical recurrence for OEIS A252617, proved by transfer matrix

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## Abstract

OEIS A252617 counts  $4 \times (n+2)$  arrays over  $\{0, \dots, 3\}$  subject to a condition imposed on every  $3 \times 3$  subblock, and it carries an empirical recurrence of order 49 contributed by R. H. Hardin. It is true, and it is not an empirical matter. A  $3 \times 3$  subblock spans three consecutive columns, so the state of a transfer matrix is a PAIR of consecutive columns and appending one more column is a step; an array is then a walk, and  $a(n)$  is a walk count on  $S = 65536$  vertices. Such a sequence is  $C$ -finite, so whether a given recurrence annihilates it is decided by a finite exact computation, with the Cayley–Hamilton theorem bounding the work.

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## 1 The conjecture

OEIS A252617 is “Number of  $(2+2) \times (n+2)$   $0..3$  arrays with every  $3 \times 3$  subblock row and diagonal sum equal to 0 3 5 6 or 8 and every  $3 \times 3$  column and antidiagonal sum not equal to 0 3 5 6 or 8.”. It has offset 1 and begins

725, 1379, 2815, 6019, 13879, 31514, 71856, 170404, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical:  $a(n) = a(n-1) + 48*a(n-3) - 48*a(n-4) - 939*a(n-6) + 939*a(n-7) + 9891*a(n-9) - 9891*a(n-10) - 63231*a(n-12) + 63231*a(n-13) + 263827*a(n-15) - 263827*a(n-16) - 759379*a(n-18) + 759379*a(n-19) + 1583561*a(n-21) - 1583561*a(n-22) - 2493304*a(n-24) + 2493304*a(n-25) + 3037897*a(n-27) - 3037897*a(n-28) - 2885834*a(n-30) + 2885834*a(n-31) + 2133236*a(n-33) - 2133236*a(n-34) - 1202312*a(n-36) + 1202312*a(n-37) + 495108*a(n-39) - 495108*a(n-40) - 137768*a(n-42) + 137768*a(n-43) + 20352*a(n-45) - 20352*a(n-46) - 1152*a(n-48) + 1152*a(n-49)$  for  $n > 52$ .

As of the “Last modified” line on the live entry (May 27 2021 15:02:06, revision 6) this is still recorded as empirical, and nothing on the entry records it as proved.

## 2 Three consecutive lines

Write an admissible array as a sequence of columns  $u_1, u_2, \dots$ , each an element of  $\{0, \dots, 3\}^4$ . A  $3 \times 3$  subblock occupies three consecutive columns and three consecutive rows, so with  $r, s, t$  three consecutive columns the blocks it contributes are

$$g = \begin{pmatrix} r_i & s_i & t_i \\ r_{i+1} & s_{i+1} & t_{i+1} \\ r_{i+2} & s_{i+2} & t_{i+2} \end{pmatrix}$$

for each admissible index, and the entry's condition is the requirement that

every row, diagonal sum  $\in \{0, 3, 5, 6, 8\}$  and every column, antidiagonal sum  $\notin \{0, 3, 5, 6, 8\}$

holds for every such  $g$ . This involves  $r$ ,  $s$  and  $t$  only: it is a condition on three consecutive columns and on nothing else.

Take as vertices of a digraph  $G$  the ordered PAIRS  $(r, s)$  of columns, of which there are  $S = (3 + 1)^{2 \cdot 4} = 65536$ , and put an edge  $(r, s) \rightarrow (s, t)$  exactly when the condition holds for the triple  $(r, s, t)$  at every admissible index. An array with  $L$  columns is then a walk of length  $L - 2$  in  $G$ : its first two columns name the starting vertex and each further column is one step. Hence, with  $M$  the adjacency matrix of  $G$ ,

$$a(n) = \mathbf{1}^\top M^{n-1+1} \mathbf{1},$$

the exponent being fixed by the entry's own shape and checked against its published terms. In particular  $a$  is  $C$ -finite.

### 3 The proof

**Lemma 1.** *Let  $q(t) = t^{49} - \sum_i c_i t^{49-i}$  be the characteristic polynomial of the conjectured recurrence. Then the residual  $a(n) - \sum_i c_i a(n-i)$  equals  $\mathbf{1}^\top M^j q(M) \mathbf{1}$  for the corresponding walk index  $j$ , so the recurrence holds from some point on if and only if  $\mathbf{1}^\top M^j q(M) \mathbf{1} = 0$  for all large  $j$ ; and by the Cayley–Hamilton theorem it is enough to examine  $j < S$ .*

*Proof.* Factor  $M^j$  out of  $M^{j+49} - \sum_i c_i M^{j+49-i}$ . For the bound,  $M^S$  is an integer combination of  $I, M, \dots, M^{S-1}$ , so the numbers  $u_j = \mathbf{1}^\top M^j q(M) \mathbf{1}$  satisfy the monic linear recurrence given by the characteristic polynomial of  $M$ ;  $S$  consecutive zeros therefore force every later one, and the last nonzero  $u_j$  pins the threshold exactly.  $\square$

**Theorem 2.**

$$a(n) = a(n-1) + 48a(n-3) - 48a(n-4) - 939a(n-6) + 939a(n-7) + 9891a(n-9) - 9891a(n-10) - 63231a(n-12)$$

for every  $n > 52$ .

*Proof.* The vector  $w = q(M) \mathbf{1}$  was formed by 49 matrix–vector products in exact integer arithmetic and  $\mathbf{1}^\top M^j w$  evaluated for  $j = 0, 1, \dots$ , again exactly. Those integers vanish from the index corresponding to  $n = 52 + 1$  onwards, and the run was continued until the working vector was identically zero, which settles all larger  $j$  at once. By Lemma 1 the recurrence holds for every  $n > 52$ .  $\square$

### 4 Verification

Three checks, each independent of the algebra above.

First, the transfer model reproduces all 24 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on the entry's own published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix, so no rounding or sampling enters anywhere.

## References

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- [2] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
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- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.